Geospatial RAP Tool: User Manual

1. Introduction

The Geospatial RAP (Reproducible Analytical Pipeline) Tool is designed to support public health analysts and epidemiologists in exploring spatial patterns in aggregated epidemiological data.

By applying spatial statistical methods to weekly case counts, the tool generates prevalence maps and identifies geographic clustering through Local Indicators of Spatial Association (LISA). These outputs help highlight areas of unusually high or low disease activity, supporting data-driven decision-making and targeted public health interventions.

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2. Getting Started: Loading Your Data

Before running the spatial analysis, you must load your aggregated epidemiological dataset. The tool accepts CSV files generated by the API-POP pipeline.

Once uploaded, the application will automatically:

- detect and standardize column names,
- validate geographic identifiers,
- and prepare the dataset for spatial analysis.

Required Data Structure:
Your dataset should include, at minimum:

- geographic identifiers (e.g., level1, level2, level3),
- epidemiological year and week,
- number of cases,
- population values.

Geographic Levels:
The application supports multiple administrative levels (e.g., province, department, fraction). These are mapped dynamically based on the configuration file, allowing flexibility across different countries.

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3. Step-by-Step Configuration

Once your data is loaded, use the left-hand sidebar to configure the spatial analysis.

A. Geographic Selection

- **Geographic Level:**
  Select the administrative level for analysis (e.g., Province, Department, Censal Fraction).
  The application will aggregate data to match the selected level.

- **Specific Locations:**
  Use the dropdown to filter the analysis to specific regions.
  You may select one or multiple locations, or leave it empty to include all available areas.

B. Temporal Selection

- **Year:**
  Select the epidemiological year to analyze.

- **Time Mode:**
  Choose how to define the analysis period:

    - **Single Week:** Analyze one specific epidemiological week

    - **Week Range:** Aggregate data across a selected range of weeks

This allows comparison between short-term snapshots and broader temporal trends.

C. Data Aggregation

The application will automatically:

- aggregate case counts and population
- calculate prevalence (cases per 100,000 inhabitants)
- prepare the dataset for spatial analysis

No manual parameter tuning is required.

D. Run Analysis

Once all selections are configured, click the **Run Spatial Analysis** button.

The application will:

- join your data with spatial boundaries
- compute spatial statistics
- generate maps and results

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4. Reviewing the Results

After the analysis completes, use the main panel tabs to explore the outputs.

Prevalence Map Tab

Displays a choropleth map representing disease prevalence across geographic units.

- Darker colors indicate higher prevalence
- Lighter colors indicate lower prevalence

This map provides a quick overview of spatial distribution.

LISA Cluster Map Tab

Displays spatial clustering results using Local Moran's I.

Each area is classified as:

- **High-High:** High values surrounded by high values (hotspots)
- **Low-Low:** Low values surrounded by low values (cold spots)
- **High-Low:** High values surrounded by low values (spatial outliers)
- **Low-High:** Low values surrounded by high values (inverse outliers)
- **Not Significant:** No statistically meaningful clustering

This helps identify areas of concern or unusual patterns.

Results Table Tab

Provides a detailed table containing:

- prevalence values
- cluster classification
- spatial identifiers

Use the **Download CSV** option to export results for further analysis.